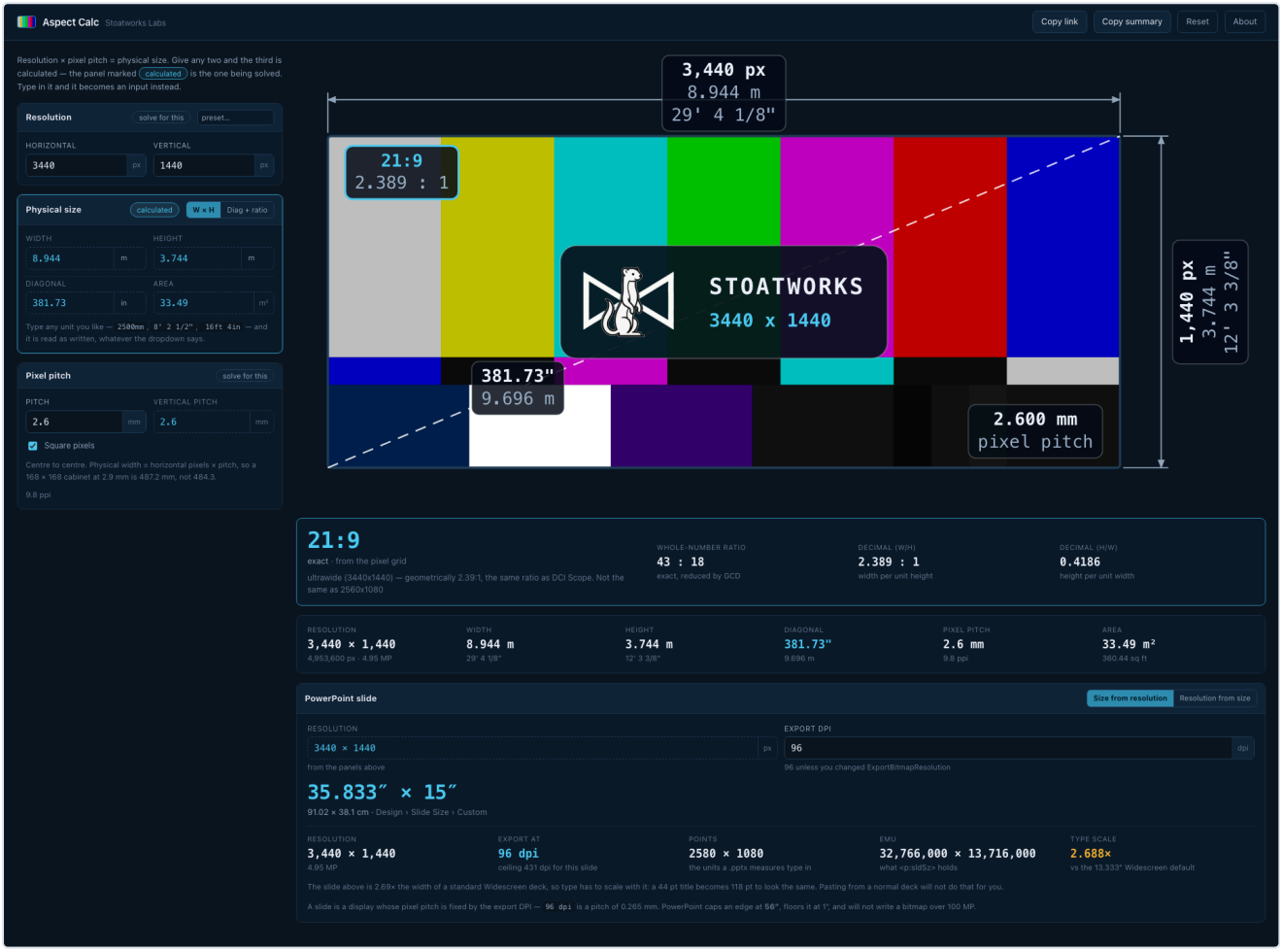
Aspect Calc user guide

Aspect Calc is an aspect ratio, pixel pitch and display geometry calculator for people who put pictures on walls. It runs in the browser only, with no account and no backend, and nothing you type leaves the tab.



3440 × 1440 at a 2.6 mm pitch: an 8.944 m wall, the ratio named 21:9 with its exact 43:18 beside it, the display drawn to scale with every dimension on it, and the PowerPoint slide size (35.833″ × 15″) that exports to that resolution.

The whole model is one relation:

```
resolution (px)  ×  pixel pitch (mm)  =  physical size (mm)
```

Give it any two and it works out the third. Diagonal and aspect ratio are two more ways of writing the physical size, so you can also hand it a 55" diagonal and "16:9" and get the width and height back.

> **Before you rely on this:** the arithmetic is pinned by 100 tests, including the ratio-naming rules, which are the only part with any judgement in them.
>
> **Nothing here has been checked by measuring a real screen**, because there's nothing to measure: it's arithmetic over the numbers you type, and it's only as right as they are.
>
> This codebase was created with AI assistance, directed and reviewed by a human author.

What it does

- **Aspect ratio from anything:** a resolution in pixels, a physical size in mm/cm/m, inches or feet-and-inches, or a diagonal and a ratio.
- **Pixel pitch in any direction:** pitch from resolution and size, size from resolution and pitch, or resolution from size and pitch. Non-square pixels are reported, not averaged away.
- **Both decimal forms**, `1.778 : 1` and `0.5625`.
- **Diagonal and area**, always, in both measuring systems at once.
- **PowerPoint slide sizes**, both directions.
- **The display drawn to scale**, with colour bars and every dimension on it.

It names the ratio properly

3440×1440 reduces exactly to **43:18**, and nobody has ever ordered a 43:18 monitor; it's called **21:9**. 1366×768 reduces to 683:384, which isn't a ratio so much as an accusation. The useful answer is **16:9, 0.05% off**.

Both the trade name and the arithmetic are always shown, and a match that isn't exact says so and says by how much. That last part matters, because a ratio quoted as exact when it's 0.05% off is how a picture ends up one pixel short across a wall.

Three things from the trade

Pitch is centre-to-centre and every pixel owns a full cell, so physical width is `horizontal pixels × pitch`, **not** `(pixels - 1) × pitch`.

A 168 × 168 px cabinet at 2.9 mm is **487.2 mm** square, not 484.3. Getting this wrong loses one pitch across the whole wall, which is invisible in a spreadsheet and very visible when the last cabinet doesn't fit the frame.

The two 21:9s are not the same ratio. 2560×1080 is 64:27 (2.370:1); 3440×1440 is 43:18 (2.389:1). They're 0.8% apart and both are sold as "21:9". The calculator names them both 21:9 and then tells you which one you have.

The colour bars are a picture, not a signal. The pattern is stretched to whatever aspect you're looking at (which is what a real generator does), and the RGB values are the standard 75% bar values but aren't colour-managed. **Don't grade against it.**

PowerPoint

Both directions: the slide size for a target resolution, or the resolution a given slide size exports to.

It handles the format's own limits (the **56-inch slide cap**, the 1-inch floor and the 100 MP export ceiling) and gives you points and EMU alongside inches and centimetres, because those are the units the file format actually stores.

If a number looks wrong

A ratio is named 16:9 but the numbers don't reduce to it. That's on purpose: the trade name and the exact reduction are both shown, with the error between them.

A wall comes out one pitch too small. You're using `(pixels - 1) × pitch`. See above.

The pitch differs horizontally and vertically. The pixels aren't square, and the tool reports that rather than averaging it away. It's a real property of some LED products.

PowerPoint refuses the slide size. It's past the 56-inch cap. The calculator says so rather than producing a number the format can't hold.
